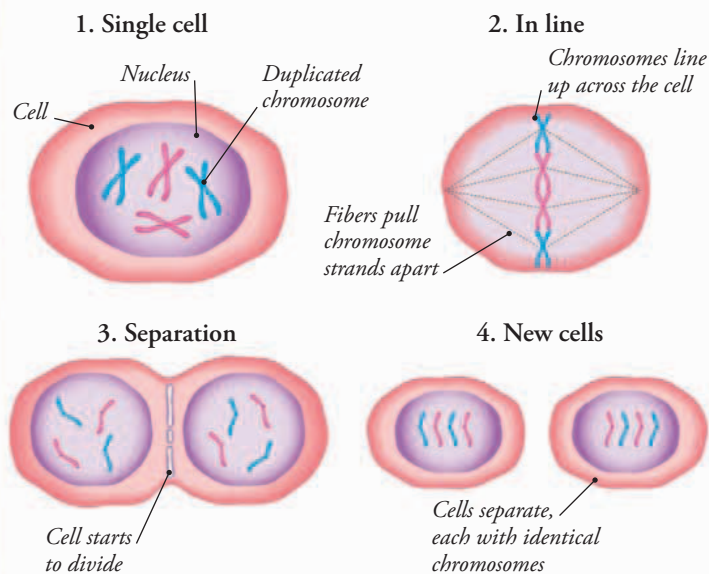




A cell membrane surrounds and protects the stem cell, holding the contents together.

MAKING NEW CELLS

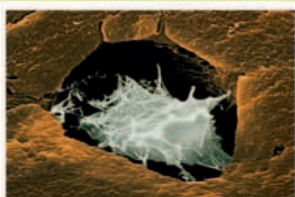
Cells divide repeatedly to replace worn-out cells—a process known as mitosis. Before a cell divides, chromosomes, which carry the instructions needed to build and run a cell, copy themselves within the nucleus (control center) of the cell. Then the cell splits into two identical cells, each with a full set of instructions.



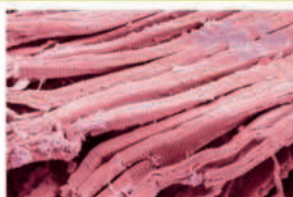
WOW!

The human body is built from 100 trillion cells and produces about 5 million cells every second to replace those that die.

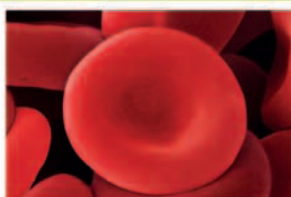
▲ **STEM CELL** *We start life as a tiny embryo growing inside our mother. An embryo contains stem cells like this one that can develop into any one of the cells found inside the body.*



▲ **BONE CELL**
This bone cell (osteocyte) maintains the bone tissue (brown) that surrounds it.



▲ **MUSCLE CELLS**
All movement is produced by muscle cells. These cardiac muscle cells make the heart beat.



▲ **RED BLOOD CELLS**
These small, dimpled blood cells carry oxygen from the lungs to body tissues.



SEEING SMALL

The invention of the microscope in the 1600s opened up a hidden world for English scientist Robert Hooke. He devised a microscope (shown above) to study tiny objects such as plants and insects. In 1665, Hooke recorded his findings in his groundbreaking book, *Micrographia* (meaning "small drawings"). He also invented the term *cell* to describe the basic unit of life.